ROOTS MINI APP

Delivery × Stock state map

Generated 2026-05-19 · internal reference

Two independent axes drive every customer outcome: **(1) what zone matched their address** and **(2) what delivery slot they picked**. Stock-checking is a third lever that's off unless both axes line up just right. This document maps every combination so nothing is missed.

A · The 4 zone outcomes

Customer address gets matched to one of these four buckets at /checkout/quote time. The free-text fallback (deployed 2026-05-18) handles WeChat-imported addresses where district is blank but the text contains the district name.

Zone	Match condition	Detected by	What the customer sees
A1. GZ district — time-slot mode	zone_type='gz_district' AND day_only=0	structured address.district, OR free-text scan for ■■■ / ■■■ / ■■■ etc.	Today's 2-hour chips (1–3pm, 3–5pm…) + 'Schedule for later' calendar
A2. GZ district — day-only mode	zone_type='gz_district' AND day_only=1	same as A1	'Express (1–2 days)' / 'Get it today (+¥X)' if same-day on / 'Schedule for later' calendar
A3. Other city	zone_type='city'	address.city exact, OR free-text scan	'Schedule for later' calendar only (no today)
A4. No match	nothing active matches	both above failed	Red 'Sorry, we don't deliver here' banner + Place Order disabled

If you ever see "no match" for a known GZ address, the district name probably isn't in the table yet — that's the only thing that breaks A1/A2 matching.

B · The 4 delivery slot shapes

What the mini-app sends as delivery_slot_id when the customer presses Place Order, and which zones can ever surface each.

slot_id sent	Origin in the UI	Same-day ?	Available in
'right_away'	'■■■■ / Right away' radio — only when zone allows + kitchen open + no surcharge	✓	A1 only (suppressed when zone has surcharge)
'YYYY-MM-DDTHH:MM' ***today***	Tap a today time-slot chip (1–3pm etc.)	✓	A1

slot_id sent	Origin in the UI	Same-day ?	Available in
'YYYY-MM-DD' ***today***	'Get it today (+¥X)' day-chip	✓	A2 only (when same-day on AND surcharge > 0)
'YYYY-MM-DD' ***future*** **(≥ today+3)**	'Schedule for later' calendar picker	—	A1 / A2 / A3
'YYYY-MM-DD' **= schedule.min_date**	'Express delivery (X+ days)' radio (synthesizes min_date pick)	—	A2 only

C · Stock-check truth table

The canonical predicate, set once at the top of both /checkout/quote and POST /orders:

```
requiresStockGate = isGuangzhou(address) && isSameDay(slot)
```

Zone	Slot picked	Stock check?	Soft-reserve at order time?
A1 (GZ time-slot)	Right_away or today chip	**YES**	**YES**
A1 (GZ time-slot)	Future-date calendar	NO	NO
A2 (GZ day-only)	Today day-chip	**YES**	**YES**
A2 (GZ day-only)	Express (= min_date, future)	NO	NO
A2 (GZ day-only)	Schedule for later	NO	NO
A3 (other city)	any slot	NO	NO
A4 (no match)	—	—	order refused before stock ever checked

D · When the stock check IS on — what the customer can see

Only fires when the gate above is YES. Then analyzeShortage runs and the modal path is:

Nearest can fulfill?	Fallback can fulfill?	District in GZ Express excluded list?	Customer sees
✓	—	—	Nothing — order proceeds silently.
✗	✓	✓ excluded	**simple_confirm** modal — 'We're out at nearest; ship from farther store? Yes / Cancel'
✗	✓	not excluded	**5-option modal** — Yes ship from fallback / Wait for restock / Cancel missing items / Replace with substitutes / Cancel order
✗	✗	—	**insufficient_stock** error — order refused at place-order time
only 1 store	(single store + out)	—	**insufficient_stock** error

E · What stays untouched by the stock gate

These rails run regardless of the matrix above:

- **Delivery fee** — always computed from the zone (base_fee, max_fee, free_threshold).
- **Packing fee** — always computed from zone (city zones only; qty × per_item, capped).
- **Same-day surcharge** — added when the picked slot is same-day in a zone with same_day_surcharge > 0, regardless of stock.
- **Voucher** — applies the same.
- **Points earn / redeem** — same.

- **Zone-not-matched (A4)** — refuses the order before any stock logic runs.
- **Address validation** — phone / name / city / district / address all required before quote.

F · Edge cases you should know about

1. Customer picks today, then kitchen closes

If the pick was for today and the customer is slow to pay, by the time the order hits the server the kitchen is closed → resolveDeliverySlot rejects with slot_unavailable. They see an error toast and have to re-pick. No automatic re-prompt — they refresh.

2. Same-day with surcharge, A1 (Tianhe)

right_away is suppressed when zone surcharge > 0 (only today's 2h chips remain). If you turn surcharge on for Tianhe, customers lose the '■■■■' option and pay surcharge on every today chip. Flag this if it surprises you.

3. Schedule-for-later starts at day 3 always

Hardcoded SCHEDULE_MIN_LEAD = 3 in computeDeliverySlots. Even if a zone has min_lead_days = 0, the calendar starts day 3 because days 0–2 are covered by today/express. Day-2-only zones (no same-day, very fast restock) aren't expressible right now.

4. City zones never offer today

No today_slots or day-chip path for zone_type='city'. Even if a city zone had same_day_allowed=1 it would never appear. Matches the rule 'outside GZ, stock isn't the bottleneck — kitchen ships when ready.'

5. Reserved stock + non-same-day order

Non-gated orders don't reserve. Two non-same-day Liwan orders for the same out-of-stock item BOTH succeed, and the office sees 2× backlog. Intentional — the kitchen restocks and fulfils — but no 'we're already overbooked next Tuesday' warning exists yet.

6. fallback_choice set on a non-gated order

If the mini-app somehow sent replace_missing or wait_for_restock on an A3/scheduled-A1 order, analyzeShortage returns gate='none' and the server refuses with no_shortage. Defensive — but a malformed client could trip this. Currently impossible since the modal that sets fallback_choice never opens.

7. Address with no district but inside Guangzhou

Free-text scan picks up '■■' if it's in the address blob. If the customer typed only 'Tianhe in English' or a colloquial form not configured as a zone name, matching fails → A4 zone-not-found. To extend, add the spelling as a name_en alias on the zone (or duplicate the district).

8. Flash-sale items + non-same-day order

Flash stock_qty IS still enforced (separate transaction inside the order POST, before the soft-reserve skip). So a flash item that's sold out can't be ordered even on a future-date Liwan order. Different rail from regular stock — intentional but worth knowing.

9. accept_fallback: true from the mini-app on a non-gated order

Harmless; routeOrder uses ignoreStock: true and ignores acceptFallback when picking the store.

G · Things this map does NOT yet cover

Open product questions:

• **Refund flow ↔ stock** — when an order is refunded, does the soft-reserve come back? releaseOrderReservations is called by the refund path for paid orders, but only if a reservation existed.

Non-gated orders had no reservation → nothing to release. Correct.

• **Awaiting-restock orders** — only created via wait_for_restock fallback choice, which only appears on gated orders. So this branch is never reached for non-GZ or scheduled orders. Consistent.

• **Wallet-only payment + future date** — works the same as any other payment method; the gate doesn't care about payment method.

> *If you want any of the edge cases above changed (e.g. **F-3** — allow day-2 calendar start; or **F-5** — show backlog warning for scheduled orders), tell us which and we'll wire them up. Otherwise this matrix is the full surface area.*